# Ashwin K M

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## Papers

- **Learn–Master–Teach Tuning: From Pupil to Professor** *Ashwin KM et al.*
  A Human-Pedagogy Inspired Fine-Tuning Paradigm for LLMs
- **FOCAL: Decoupling Conflicting Objectives in LLM Domain Adaptation via Temporal Orthogonalization** *Ashwin KM et al.*

## Education

- **Indian Institute of Science, Bangalore** *TGPA: 8.3*
  *B.Tech in Mathematics and Computing* 2023 – 2027

## Research Experience

- **Zenteiq AI**
  *ML Research Engineer* May 2025 – Dec 2025
  - Investigated whether training LLMs through a human-inspired **Learn–Master–Teach** lifecycle improves mathematical reasoning accuracy beyond static fine-tuning.
  - Designed a multi-stage training pipeline combining curriculum-grounded learning, interactive concept refinement, feedback-driven practice, multi-teacher distillation, self-reflection, hallucination mitigation, and pedagogical reinforcement learning to systematically improve reasoning reliability.
  - Found that the integrated pipeline yields **+10–17 pp accuracy gains** over the base model on CBSE-style math benchmarks, while revealing that feedback-driven refinement and teaching-oriented objectives are critical for stable gains beyond knowledge injection alone.
- **Energy Informatics Lab, Indian Institute of Science (IISc)**
  *ML Research Intern* May 2025 – Jul 2025
  - Designed a **controlled synthetic anomaly injection framework** for electricity consumption time-series data, implementing NNG-Mix–inspired Type 2/3/4 anomaly mechanisms (with emphasis on extreme negative deviations) to systematically perturb normal signals while preserving temporal structure.
  - Built a **reproducible benchmarking pipeline** for evaluating classical anomaly detectors (IQR, LOF, Isolation Forest, Modified Z-Score), integrating Kalman filtering and seasonal decomposition for imputation and trend extraction, and visualizing detection behavior via anomaly heatmaps and score trajectories.
  - Empirically analyzed detector sensitivity under **controlled anomaly regimes**, showing how anomaly type and magnitude affect false positives and miss rates, and highlighting limitations of statistical detectors when anomalies overlap with seasonal or trend components rather than manifesting as point outliers.

## Selected Projects

- **FOCAL** *Jan 2026 - Apr 2026*
  *Supervised by Prof. Shalabh Bhatnagar*
  - Current AI scientists fail to identify meaningful research directions for scientific discovery in domains with limited prior work as they don't build foundational understanding nor iteratively refine their hypothesis.
  - Proposed a multi-agent framework that acquires prerequisite knowledge, generates and critiques research hypotheses, and designs experiments to evaluate promising research directions.
  - Demonstrated up to 13.2 percentage-point improvements over ABMLL across common-sense, mathematical, coding, and medical reasoning benchmarks, while exhibiting stronger out-of-domain generalization than competing meta-learning approaches.
- **Latent Diffusion Research Agents** *Jan 2026 - May 2026*
  *Supervised by Prof. Y. Narahari*
  - Current deep research agents rely on repeated text-mediated communication between agents, introducing cumulative information loss, vocabulary bottlenecks, and increasing inference costs across long-horizon research workflows.
  - Developed a latent-space collaboration framework that replaces internal text exchanges with KV-cache transfer, localizes decoding to tool-interaction boundaries, and provides theoretical guarantees on convergence, information preservation, and iterative agent refinement.
  - Matched the performance of text-mediated baselines while reducing generated tokens by 33% and wall-clock execution time by 28%, validating latent communication as a more efficient mechanism for multi-agent coordination.
- **Benchmarking Offline Reinforcement Learning Algorithms** *Jan 2025 - April 2025*
  *Supervised by Prof. Chiranjib Bhattacharya*

- Investigated the robustness of standard offline reinforcement learning algorithms (Decision Transformer, Implicit Q-Learning, TD3-BC, AWAC, ReBRAC, Behavior Cloning) by benchmarking them on MuJoCo continuous-control tasks using the **Minari dataset format**, motivated by the need to move beyond legacy D4RL pipelines and evaluate algorithms under modern, reproducible offline-RL data interfaces.
- Refactored and rebuilt the offline RL training and evaluation pipeline to support **Minari + MuJoCo environments**, addressing incompatibilities with D4RL assumptions (dataset APIs, environment registration, trajectory handling), and ensuring fair, like-for-like comparisons across algorithms under a unified experimental setup.
- Conducted an architectural ablation replacing standard `nn.Linear` layers with `NdLinear` layers across policies and critics, observing marginal performance differences in single-dimensional settings but identifying a key limitation of the benchmarknamely that NdLinears advantages are likely to emerge in higher-dimensional tensor structureshighlighting a concrete direction for future multi-modal or structured-state offline RL evaluation.

- **IssueOps: Intelligent Repository Management** *Aug 2025 - Dec 2025*
  *Supervised by Prof. Yogesh Simhan*
  - Investigated whether a **multi-agent, event-driven automation pipeline** can reduce issue triage latency and maintainer burnout in large GitHub repositories, motivated by evidence that developers lose **~7.3 hours/week** to triage and face **3+ day delays** before issues are categorized or routed in high-noise repositories.
  - Designed and implemented a **fully serverless, asynchronous architecture** (AWS Lambda, EventBridge, DynamoDB) comprising *Intake, Triage, Assignment, and SLA agents*; integrated **LLM-based classification** (Llama-3 for spam filtering and metadata cleanup, Mistral-7B for issue type/priority/component inference) and **availability-aware routing** by inferring contributor expertise from commit history and working hours from activity timestamps.
  - Validated **end-to-end automation on live repositories**, processing batches of issues in **2-5 minutes per repository** and operating in parallel across repos; peer testing revealed accuracy limitations of a tuned BERT classifier despite >85% test accuracy, motivating a switch to **Mistral-7B via AWS Bedrock**, which improved real-world labeling reliability and highlighted trade-offs around cold starts and per-issue inference cost.

## SKILLS

- **Programming & Core ML:** Python, PyTorch, HuggingFace, LLM Routing, Multi-Fidelity Systems
- **LLMs & Agentic Systems:** PEFT, LangChain, LangGraph, vLLM, Unsloth
- **Reinforcement Learning:** Multi-Armed Bandits, TRL, OpenRLHF, Gymnasium, MuJoCo
- **Systems & Infrastructure:** Efficient Inference, AWS Cloud, Ray, Docker, Git, Weights & Biases

## ACHIEVEMENTS

- **World Scholars Cup, Global Round (Manila)** - Placed **30th out of 1,000 teams** worldwide; won **8 Gold and 5 Silver medals**; qualified for the **Tournament of Champions at Yale University** *2019*
- **National Talent Search Examination (NTSE)** - Stage I Awardee with **Rank 55** *2021*
- **Kishore Vaigyanik Protsahan Yojana (KVPY)** - Secured **Top 2000 rank** in SA stream *2021*

## POSITIONS OF RESPONSIBILITY

- **Core Coordinator Nexus (SciTech Fest), Rhapsody 2025, IISc** *Dec 2024 – Mar 2025*
  - Founded and led **Nexus**, scaling it into a flagship SciTech platform at IISc focused on **AI, Robotics, IoT, and Automation**.
  - Drove **industry partnerships and large-scale competitions**, collaborating with **Autodesk, L&T, OLA, ART-PARK** and spearheading **Robonautica, Code Conquest, InnovateX**.
- **Sponsorship Coordinator Pravega XI, IISc** *Aug 2024 – Dec 2024*
  - Led **sponsorship acquisition and corporate outreach**, securing substantial funding and long-term industry partnerships.
  - Aligned **corporate branding with Pravegas science and technology mission**, strengthening sponsor retention and studentindustry collaboration.
- **Event Coordinator RevCoding, Pravega 2025, IISc** *Jul 2024 – Jan 2025*
  - Conceptualized and organized **RevCoding**, a reverse-coding competition integrating **algorithmic reasoning, logical deduction, and narrative-driven problem solving**.
  - Designed multi-stage challenges requiring participants to decode hidden clues across code-based problem settings.